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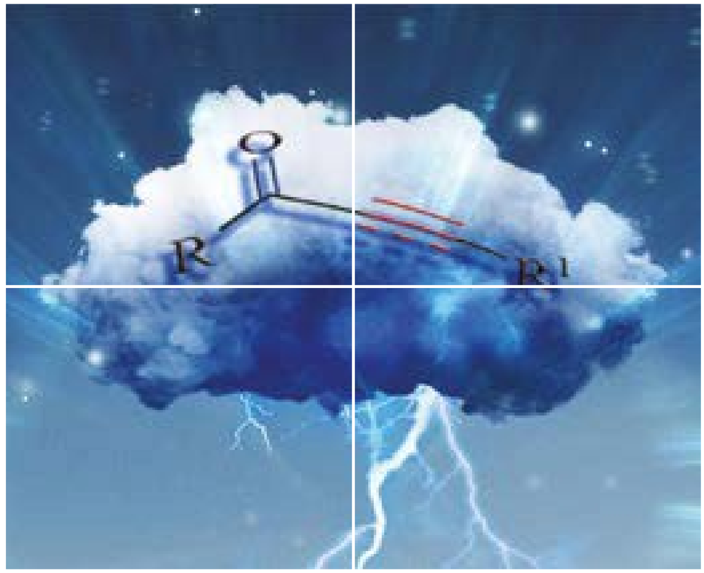

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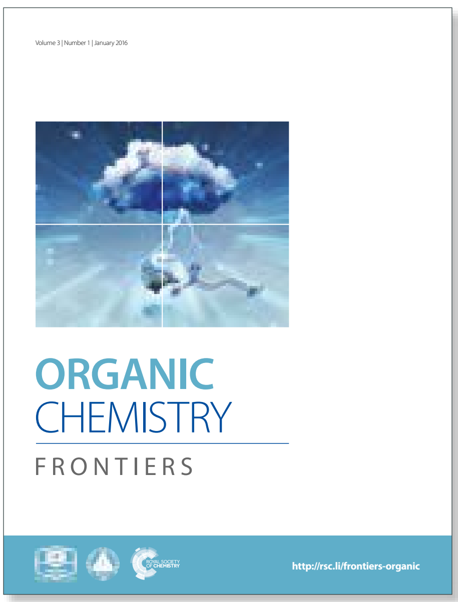

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Journal Name

COMMUNICATION

Cobalt-Catalyzed Asymmetric Reactions of Heterobicyclic Alkenes with *in situ* Generated Organozinc Halides

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A general strategy for the asymmetric allylation and benzylation of heterobicyclic alkenes were described by employing the *in situ* generated organozinc halides. The methodology features the application of a co-catalytic system comprising chiral cobalt complex and Lewis acid, delivers both the asymmetric ring opening products of oxabenzonorbornadienes and the asymmetric addition products of azabenzonorbornadienes respectively.

Organozinc reagents are frequently used organometal nucleophiles in organic synthesis as they have provided efficient method for the formation of new C-C bonds. Being as soft nucleophile, organozinc regent possesses the advantages of wide functional group tolerance and improved chemoselectivity compared to organolithium or Grignard reagent.[1] Accordingly, a number of name reactions have been developed by employing organozinc reagents, such as Negishi coupling,[2] Fukuyama coupling,[3] Reformatsky reaction,[4] Simmons-Smith reaction,[5] and Barbier reaction[6].

On the other hand, the desymmetrization of heterobicyclic alkenes has resulted into the successful synthesis of various cyclic products with high enantioselectivities.[7] Additionally, some of these reactions have been used as key steps in the synthesis of natural or bioactive molecules which has been illustrated by the group of Lautens.[8] Despite the great progress made in this field, noble metals such as rhodium, palladium, and iridium are frequently required.[9] Although remarkable progress has been made by using of nickel, cobalt, copper and iron catalysts,[10] it is still highly desirable to explore non-precious metal catalytic system with extended reaction scope and enhanced enantioselectivity of the reaction. Our group has a continuous interest in the asymmetric reactions of bicyclic alkenes, and we has proved that the transitional metal/Lewis acid co-catalytic system was in particularly suitable for the asymmetric ring opening reactions of heterobicyclic alkenes.[11] Therefore, we anticipated that the combination of chiral cobalt complex with Lewis acid could lead to outstanding result in the asymmetric reactions of heterobicyclic alkenes with organozinc nucleophiles.

Inspired by the Barbier reaction, which describes the nucleophilic reaction of a carbonyl compound with organozinc reagents generated *in situ*,[6] we envisioned that the asymmetric reaction of heterobicyclic alkenes could be realized with the *in situ* generated organozinc halides. Initially, the reaction of oxabenzonorbornadiene **1a** with allyl bromide **2b** was investigated by using chiral cobalt catalyst comprising $CoCl_2$ and a chiral ligand. As the experimental results outlined in table 1, initially tested (*R*,*R*)-Binap resulted to naphthalen-1-ol and 2-allylnaphthalene, but no desired product was detected. By using (*R*)-Segphos, **3aa** was obtained in 34% yield with low enantioselectivity. Switching to bidentate phosphines that bearing point chirality, it was pleased to find that (*R*,*R*)-BDPP gave the anticipated product **3aa** in 74% yield with 85% *ee*. Although good yield was achieved by (*R*,*R*)-Chiraphos, the *ee* was low. Bisphospholanoethane such as (*R*,*R*)-Me-BPE and (*R*,*R*)-Ph-BPE were also not suitable chiral ligands in term of the yields of reactions. The absolute configuration of product **3aa** was assigned by comparison of the chiral HPLC data with that reported in the literature.[10c]

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Electronic Supplementary Information (ESI) available: Experimental procedures, product characterization, copies of HPLC, ^{1}H and ^{13}C NMR spectrums, X-ray analysis of **3hf**. See DOI: 10.1039/x0xx00000x

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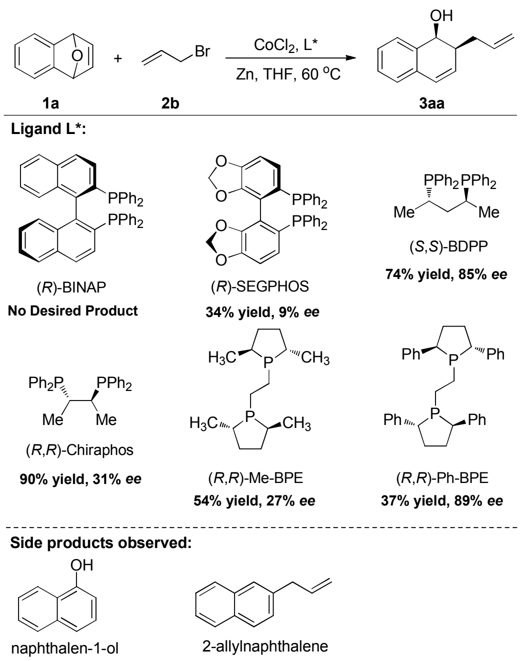
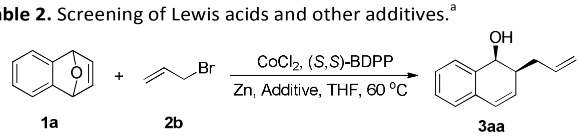


Table 1. Initial ligand screening of the asymmetric ring opening reactions of oxabenzonorbornadiene with allyl bromide. Reaction conditions: **1a** (0.2 mmol), **2b** (1.0 mmol), $CoCl_2$ (0.02 mmol), chiral bidentate ligand (0.024 mmol), and Zinc powder (1.0 mmol) in tetrahydrofuran (2 mL) at 60 °C for 1 h under an argon atmosphere, isolated yields, *ee*s were determined by chiral HPLC.

Table 2. Screening of Lewis acids and other additives.[a]

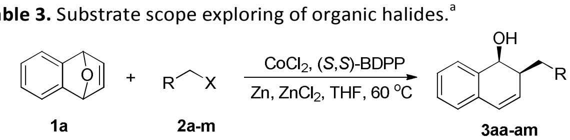


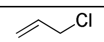
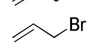
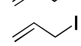
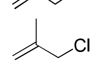
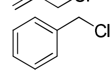
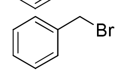
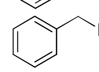
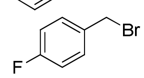
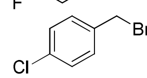
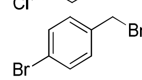
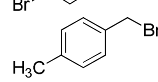
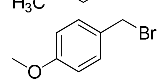
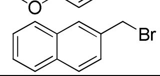
Entry	Additive	Time [h]	Yield [%][b]	*Ee* [%][c]
1	---	1	74	85
2	$Zn(OTf)_2$	1	59	55
3	$Al(OTf)_3$	1	74	58
4	$Fe(OTf)_2$	1	61	73
5	$Cu(OTf)_2$	1	45	9
6	$ZnCl_2$	1	81	90
7	$ZnBr_2$	1	54	85
8	ZnI_2	1	0	-
9	CuCl	3	16	45
10	$FeCl_2$	3	71	87
11	$AlCl_3$	1	47	69
12	$^tBu_4NCl$	1	56	80

[a] Reaction conditions: The reaction was carried out with **1a** (0.2 mmol), **2b** (1.0 mmol), $CoCl_2$ (0.02 mmol), (*S*,*S*)-BDPP (0.024 mmol), additive (0.1 mmol), and Zinc powder (1.0 mmol) in tetrahydrofuran (2 mL) at 60 °C under an argon atmosphere for the time indicated.
[b] Isolated yield of product **3aa**.
[c] Determined by HPLC with a Chiralcel OD-H column.

Subsequently, effort was dedicated to identify the addition of a suitable Lewis acid as co-catalyst (Table 2). We quickly learned that the trifluoromethanesulfonate salts were not suitable, as decreased *ee*s were observed (Table 2, entries 2-5). By switching to some halogen salts, it was delighted to find the using of $ZnCl_2$ gave the desired product in 81% yield with 90% *ee* (Table 2, entry 6). The next tested $ZnBr_2$ reduced both the yield and enantioselectivity, and ZnI_2 only afforded naphthalen-1-ol as product (Table 2, entries 7-8). By assuming the chloride ion has a specific effect, other metal chlorides were studied, but none were effective compared to $ZnCl_2$ (Table 2, entries 9-11). The addition of $^tBu_4NCl$ also gave reduced yield and *ee* (Table 2, entry 12). In all cases of low yields of the ring opening reactions, naphthalen-1-ol was generated as by-product from oxabenzonorbornadiene **1a** directly and could not be completely suppressed.

Table 3. Substrate scope exploring of organic halides.[a]



Entry	Organohalide	Time [h]	Yield [%][b]	*Ee* [%][c]
1	 **2a**	3	86	97
2	 **2b**	1	81	90
3	 **2c**	0.5	64	84
4	 **2d**	9	19	92
5	 **2e**	13	NR	---
6	 **2f**	0.5	85	97
7	 **2g**	0.5	58	94
8	 **2h**	0.5	88	97
9	 **2i**	0.5	87	96
10	 **2j**	0.5	78	97
11	 **2k**	0.5	87	96
12	 **2l**	0.5	56	96
13	 **2m**	0.5	71	96

[a] Reaction conditions: The reaction was carried out with **1a** (0.2 mmol), **2** (1.0 mmol), $CoCl_2$ (0.02 mmol), (*S*,*S*)-BDPP (0.024 mmol), $ZnCl_2$ (0.1 mmol), and Zinc powder (1.0 mmol) in tetrahydrofuran (2 mL) at 60 °C under an argon atmosphere for the time indicated.
[b] Isolated yield of product **3**.
[c] Determined by HPLC with a Chiralpak AD-H or Chiralcel OD-H column.

The scope of the current reaction was surveyed by employing a wide range of organic halides. As the experimental results summarized in table 3, allyl chloride, allyl bromide, and allyl iodide are all viable substrates to afford the

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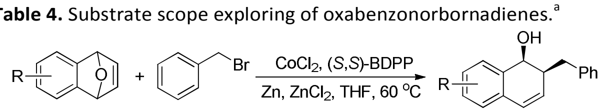
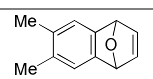
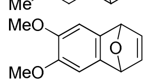
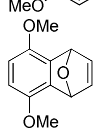
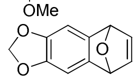
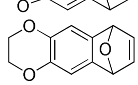
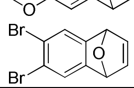
desired products (Table 3, entries 1-3). And it has been observed that allyl iodide showed highly activity but gave inferior yield and enantioselectivity (Table 3, entry 3). Although the β-branched allyl chloride was applicable to give a good *ee*, the reaction yield was very low due to the generation of naphthalen-1-ol as by-product (Table 3, entry 4). Benzyl chloride could not afford any desired product, as the generation of organozinc from benzyl chloride directly was not feasible due to its inactive property (Table 3, entry 5).[12] To our delight, the more reactive benzyl bromide delivered the ring opening product in good yield with a high *ee* of 97% (Table 3, entry 6). Benzyl iodide also participate readily to give excellent enantioselectivity albeit with reduced yield caused by the generation of naphthalen-1-ol (Table 3, entry 7). Benzyl bromide with halogen substituents at *para*-position were converted to the desired product smoothly, leaving the halogen groups group intact (Table 3, entries 8-10). And the electron-rich substrates such 4-methylbenzyl bromide and 4-methoxybenzyl bromide were well tolerate (Table 3, entries 11-12). In addition, 2-(bromomethyl)naphthalene was also reactive in the present transformation (Table 3, entry 13).

Then, the scope of current ring opening reaction was further studied by employing various oxabenzonorbornadienes (Table 4). In general, satisfactory results with high levels of enantioselectivity were given, and the bromo-bearing substrate was compatible (Table 4, entries 1-6).

Table 4. Substrate scope exploring of oxabenzonorbornadienes.[a]



1b-g **2f** **3bf-gf**

Entry	1b-g	Time [h]	Yield [%][b]	*Ee* [%][c]
1	 **1b**	0.5	80	96
2	 **1c**	0.5	93	96
3	 **1d**	0.5	94	97
4	 **1e**	0.5	67	95
5	 **1f**	0.5	96	97
6	 **1g**	0.5	61	95

[a] Reaction conditions: The reaction was carried out with **1** (0.2 mmol), **2f** (1.0 mmol), $CoCl_2$ (0.02 mmol), (*S*,*S*)-BDPP (0.024 mmol), $ZnCl_2$ (0.1 mmol), and Zinc powder (1.0 mmol) in tetrahydrofuran (2 mL) at 60 °C under an argon atmosphere for the time indicated.

[b] Isolated yield of product **3**.

[c] Determined by HPLC with a Chiralpak AD-H, Chiralcel OD-H or OJ-H column.

Interestingly, the present co-catalytic system was also applicable to azabenzonorbornadiene **1h** and **1i** by extending the reaction time, afforded the addition product **3hf**[13] and **3if** with 86% *ee* instead (Figure 1, a). However, the reaction of **1h** and allyl bromide **1b** gave poor result. In order to gain a better understanding of the reaction mechanism, the reaction of azabenzonorbornadiene **1h** with benzyl bromide was quenched by deuterated acetic acid, which has delivered the deuterated addition product **3hf-D** (Figure 1, b). Judging from H-NMR analysis, the deuterium was completely incorporated into the pyrrolidine part of the product. This result indicates that a cobalt-intermediate **C** was formed and finally giving the addition product **3hf** by cation exchange (Figure 2, c). However, when oxabenzonorbornadiene was used (X = O), the intermediate **C** undergo β-elimination and rearrangement to afford the ring opening product **3af**. The divergent mechanism could be explained by the higher activity of oxabenzonorbornadiene compared with the corresponding azabenzonorbornadiene.[14]

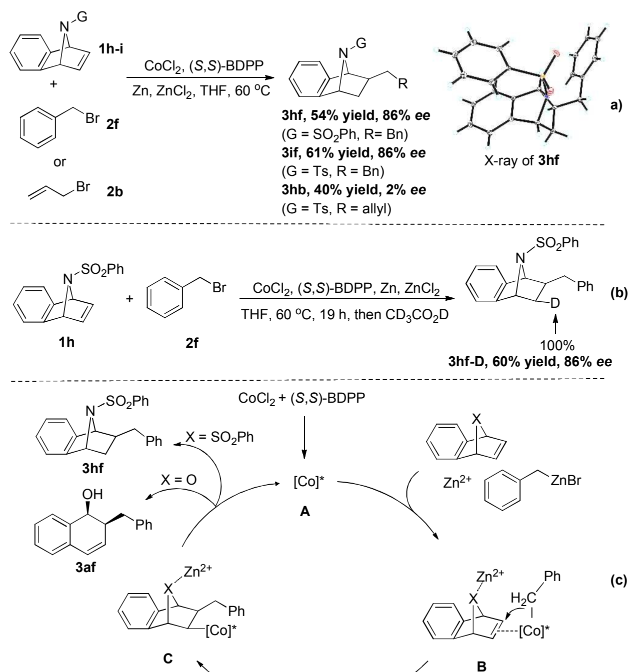


Figure 1. The reaction of azabenzonorbornadienes and the proposed reaction mechanism.

Conclusions

In conclusion, the asymmetric reactions of heterobicyclic alkenes has been investigated with *in situ* generated organozinc halides for the first time. The catalytic system comprising $CoCl_2$, (*S*,*S*)-BDPP, and $ZnCl_2$ successfully enabled the asymmetric ring opening reaction of oxabenzonorbornadienes and asymmetric addition reaction of azabenzonorbornadienes respectively. Various organozinc halides that generated from both allyl halides and benzyl

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halides with zinc powder were suitable nucleophiles and gave the corresponding chiral product in excellent enantioselectivities. Further exploration of the cobalt/zinc co-catalytic system in other asymmetric reactions are currently under way.

Experimental Section

Typical Experimental Procedure for the Asymmetric Reactions of Heterobicyclic Alkenes

Cobalt chloride (0.020 mmol), (*S*,*S*)-BDPP (0.024 mmol), zinc powder (1.0 mmol), zinc chloride (0.1 mmol) and anhydrous THF (1 mL) were added to a Schlenk tube under argon atmosphere, the resulting solution was stirred for 30 min. Then a solution of oxabenzonorbornadiene **1a** (0.2 mmol) and allyl bromide **2a** (1.0 mmol) in anhydrous THF (1 mL) was added, and the resulting solution was stirred at 60 °C under argon atmosphere with TLC monitoring until the complete consumption of **1a**. The reaction mixture was quenched by saturated ammonium chloride aqueous and extracted with diethyl ether for three times. The combined organic layers were dried over Na_2SO_4 and concentrated under reduced pressure to afford the crude product, which was purified by neutral aluminum oxide chromatography using hexanes and ethyl acetate (10:1) as eluents to give **3aa** in 86% yield. The enantiomeric excess of **3aa** was determined as 97% by chiral HPLC analysis using a Chiralpak AD-H column.

Acknowledgements

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